
ST3101: 1. Exercises on basic operations

For definitions of basic matrix operations, refer to Chapter 4 of Zelderman and ‘Matrix operations.pdf’ in the ‘Linear algebra cheat sheets’ folder.

1. Constructing matrices

For two weeks we record the TV shows that each of six families watch on Netflix:

Family 1: Breaking Bad, Lincoln Lawyers

Family 2: The Office, Seinfeld

Family 3: Peaky Blinders, Better call Saul

Family 4: The Office, Arrested Development, Seinfeld

Family 5: Beef, Lincoln Lawyers

Family 6: Black Mirror, Beef

(a) Construct an 6×9 matrix $\mathbf{A}$ of families by shows, with a cell equal to 1 if a family watches the show, and zero otherwise (you can leave 0 cells empty).

(b) Write the number of shows that both families 5 and 6 in terms of the inner product of vectors.

(c) If $\mathbf{a}_i$ is the i ’th column of $\mathbf{A}$, explain in words what

$$\mathbf{a}_i^\top \mathbf{a}_i$$

is.

(d) For columns i and j , corresponding to ‘The Office’ and ‘Seinfeld’, respectively, explain in words what

$$\mathbf{a}_i^\top \mathbf{a}_j$$

is.

(a) We let the columns correspond to the 9 unique shows

1	2	3	4	5	6	7	8	9
SF	TO	PB	BB	LL	BCS	B	AD	BM

and the rows correspond to each of the six families so that $a_{ij} = 1$ if family i watches show j , $i = 1, \dots, 6$, $j = 1, \dots, 9$. Our matrix is then

$$\mathbf{A} = \begin{bmatrix} & & & & & 1 & 1 & & \\ 1 & 1 & & & & & & & \\ & & 1 & & & & 1 & & \\ 1 & 1 & & & & & & & 1 \\ & & & & & 1 & & 1 & \\ & & & & & & & 1 & 1 \end{bmatrix}$$

(b) The number of shows that family $i = 5$ and $j = 6$ both watch is the number of k such that $a_{ik} = a_{jk} = 1$, in other words k such that $a_{ik}a_{jk} = 1$. To sum these up, write

$$\sum_{k=1}^n a_{ik}a_{jk}$$

which we recognise as the inner product

$$\mathbf{b}_i \mathbf{b}_j^\top$$

if we define

$$\mathbf{A} = \begin{bmatrix} \text{---} & \mathbf{b}_1 & \text{---} \\ \text{---} & \mathbf{b}_2 & \text{---} \\ & \vdots & \\ \text{---} & \mathbf{b}_6 & \text{---} \end{bmatrix}$$

Here

$$\mathbf{b}_5 \mathbf{b}_6^\top = \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 1 \end{bmatrix} = 1$$

(c) Define $\mathbf{A}$ in terms of its columns

$$\mathbf{A} = \begin{bmatrix} \left| \right. & \left| \right. & \dots & \left| \right. \\ \mathbf{a}_1 & \mathbf{a}_2 & \dots & \mathbf{a}_9 \\ \left| \right. & \left| \right. & \dots & \left| \right. \end{bmatrix}$$

then the product

$$\mathbf{a}_i^\top \mathbf{a}_i = \sum_{j=1}^6 a_{ij}^2$$

is a sum over families. Here a_{ij} is either 1 or 0, so $a_{ij}^2 = 1$ if $a_{ij} = 1$, and zero otherwise. Consequently

$$\mathbf{a}_i^\top \mathbf{a}_i = \sum_{j=1}^6 a_{ij}$$

simply counts how many families watch show i .

(d) If a family i watches the Office, $a_{ij} = 1$, for $j = 2$, and if they watch Seinfeld $a_{ij} = 1$, for $j = 1$. Consequently, if they watch both $a_{i1} = a_{i2} = 1$, and

$$\mathbf{a}_1^\top \mathbf{a}_2 = \sum_{i=1}^6 a_{i1} a_{i2}$$

is a sum over families. Here $a_{i1} a_{i2}$ is 1 if family i watches *both* TO and SF, and zero otherwise. Consequently $\mathbf{a}_1^\top \mathbf{a}_2$ simply counts how many families watch both shows. Here this is 2 (families 2 and 4).

2. Inner products

For two 3×1 vectors $\mathbf{x}$ and $\mathbf{y}$

(a) Write

$$x_1 y_1 + x_2 y_2 + x_3 y_3$$

in terms of a product of $\mathbf{x}$ and $\mathbf{y}$.

(b) Write

$$\sum_{i=1}^3 x_i$$

in terms of a product of $\mathbf{x}$ and another vector.

(c) Let $\mathbf{x} = (1, 2, 3)^\top$ and $\mathbf{y} = (1, 1, 1)^\top$, calculate

$$\mathbf{x} \mathbf{y}^\top$$

(a) Let $\mathbf{x} = (x_1, x_2, x_3)^\top$ and $\mathbf{y} = (y_1, y_2, y_3)^\top$ be two 3×1 column vectors, then

$$\mathbf{x}^\top \mathbf{y} = \sum_{i=1}^3 x_i y_i = x_1 y_1 + x_2 y_2 + x_3 y_3$$

(b) Let $\mathbf{x} = (x_1, x_2, x_3)^\top$ and $\mathbf{y} = (1, 1, 1)^\top$ be two 3×1 column vectors, then

$$\mathbf{x}^\top \mathbf{y} = \sum_{i=1}^3 x_i \times 1 = x_1 \times 1 + x_2 \times 1 + x_3 \times 1 = \sum_{i=1}^3 x_i$$

(c) We use fact that for $\mathbf{AB} = \mathbf{C}$, the first column of $\mathbf{C}$ is the weighted sum of the columns of $\mathbf{A}$, with the rows of the first column of $\mathbf{B}$ as weights. Here $\mathbf{A}$ only has one column, $\mathbf{x}$ and $\mathbf{B}$ only has one row, hence

$$\mathbf{x} \mathbf{y}^\top = \begin{bmatrix} | \\ \mathbf{x} \\ | \end{bmatrix} \begin{bmatrix} y_1 & y_2 & y_3 \end{bmatrix} = \begin{bmatrix} | & | & | \\ y_1 \mathbf{x} & y_2 \mathbf{x} & y_3 \mathbf{x} \\ | & | & | \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \end{bmatrix}$$

3. Multiplication with scalar

The variance covariance matrix $\mathbf{S}$ for three variables X_1, X_2, X_3 is

$$\mathbf{S} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Define the variables $Y_i = 2X_i$, for $i = 1, 2, 3$, what is the covariance matrix of the variables Y_1, Y_2, Y_3 ? Write this as (i) multiplication of a matrix by a scalar, and (ii) as addition of a matrix with a diagonal matrix.

For each variable, we have that

$$V(Y_i) = V(2X_i) = 2^2 V(X_i) = 4V(X_i)$$

This means that all the variances will be multiplied by 4. The covariance matrix can thus be written (i):

$$4\mathbf{S} = \begin{bmatrix} 8 & 0 & 0 \\ 0 & 12 & 0 \\ 0 & 0 & 2 \end{bmatrix}.$$

In this particular case, since the matrix is diagonal, we can also add 6, 3, and 1 to the variances for variables 1, 2, and 3, respectively (ii):

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 6 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 8 & 0 & 0 \\ 0 & 12 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

We can construct the new covariance using (ii) but it does not really have an interpretation. The multiplication in (i) is a little more meaningful.

4. Adding matrices

Let the matrix $\mathbf{A}$ be

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}$$

What is

$$\mathbf{A} + \mathbf{B}$$

for

(a)

$$\mathbf{B} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

(b)

$$\mathbf{B} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

(a)

$$\mathbf{A} + \mathbf{B} = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix} + \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1+1 & 2+1 \\ 3+1 & 6+1 \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 4 & 7 \end{bmatrix}$$

(b)

$$\mathbf{A} + \mathbf{B} = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix} + \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1-1 & 2 \\ 3 & 6+1 \end{bmatrix} = \begin{bmatrix} 0 & 2 \\ 3 & 7 \end{bmatrix}$$

5. Multiplying vectors

Define the two 2×1 vectors $\mathbf{x}$ and $\mathbf{y}$

$$\mathbf{x} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}, \text{ and } \mathbf{y} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

(a) Calculate

$$\mathbf{x}^\top \mathbf{y}$$

(b) Calculate

$$\mathbf{x}^\top \mathbf{x}$$

(c) Calculate

$$\mathbf{xy}^\top$$

(a)

$$\mathbf{x}^\top \mathbf{y} = 2 \times 1 + 2 \times 2 = 6$$

(b)

$$\mathbf{x}^\top \mathbf{x} = 2 \times 2 + 2 \times 2 = 8$$

(c)

$$\mathbf{xy}^\top = \begin{bmatrix} x_1 y_1 & x_1 y_2 \\ x_2 y_1 & x_2 y_2 \end{bmatrix} = \begin{bmatrix} 2 \times 1 & 2 \times 2 \\ 2 \times 1 & 2 \times 2 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 2 & 4 \end{bmatrix}$$

6. Multiplying matrices

Let the matrix $\mathbf{A}$ be

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}$$

What is

$$\mathbf{AB}$$

for

(a)

$$\mathbf{B} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

(b)

$$\mathbf{B} = \begin{bmatrix} 6 & 0 \\ 0 & 1 \end{bmatrix}$$

(c)

$$\mathbf{B} = \begin{bmatrix} -2 & 0 \\ 1 & 0 \end{bmatrix}$$

(a) For each case, we write the product as a linear combination of the columns of $\mathbf{A}$

$$\mathbf{AB} = \left[\left[\begin{array}{c|c} | & | \\ b_{11}\mathbf{a}_1 & + \quad b_{21}\mathbf{a}_2 \\ | & | \end{array} \right] \quad \left[\begin{array}{c|c} | & | \\ b_{12}\mathbf{a}_1 & + \quad b_{22}\mathbf{a}_2 \\ | & | \end{array} \right] \right]$$

And, since we are using the same matrix $\mathbf{A}$ all the time

$$\mathbf{AB} = \left[\left[b_{11} \begin{bmatrix} 1 \\ 3 \end{bmatrix} + b_{21} \begin{bmatrix} 2 \\ 6 \end{bmatrix} \right] \quad \left[b_{12} \begin{bmatrix} 1 \\ 3 \end{bmatrix} + b_{22} \begin{bmatrix} 2 \\ 6 \end{bmatrix} \right] \right]$$

For the first $\mathbf{B}$

$$\mathbf{AB} = \left[\left[0 \times \begin{bmatrix} 1 \\ 3 \end{bmatrix} + 1 \times \begin{bmatrix} 2 \\ 6 \end{bmatrix} \right] \quad \left[1 \times \begin{bmatrix} 1 \\ 3 \end{bmatrix} + 0 \times \begin{bmatrix} 2 \\ 6 \end{bmatrix} \right] \right] = \begin{bmatrix} 2 & 1 \\ 6 & 3 \end{bmatrix}$$

This matrix $\mathbf{B}$, is an example of a permutation matrix. Just by looking at it we see that it will simply reorder the columns of $\mathbf{A}$.

(b) The second matrix $\mathbf{B}$ is *diagonal* so we know that multiplying it with $\mathbf{A}$ will give us a matrix with the same columns as $\mathbf{A}$ but with the first column multiplied by 6 and the second multiplied by 1:

$$\mathbf{AB} = \left[\left[6 \times \begin{bmatrix} 1 \\ 3 \end{bmatrix} + 0 \times \begin{bmatrix} 2 \\ 6 \end{bmatrix} \right] \quad \left[0 \times \begin{bmatrix} 1 \\ 3 \end{bmatrix} + 1 \times \begin{bmatrix} 2 \\ 6 \end{bmatrix} \right] \right] = \begin{bmatrix} 6 & 2 \\ 18 & 6 \end{bmatrix}$$

(c) The last matrix $\mathbf{B}$ has the last column equal to 0 so we know that the matrix resulting from the multiplication will a last column that takes 0 times the first column of $\mathbf{A}$ and 0 times the second column of $\mathbf{A}$ and:

$$\mathbf{AB} = \left[\left[(-2) \times \begin{bmatrix} 1 \\ 3 \end{bmatrix} + 1 \times \begin{bmatrix} 2 \\ 6 \end{bmatrix} \right] \quad \left[0 \times \begin{bmatrix} 1 \\ 3 \end{bmatrix} + 0 \times \begin{bmatrix} 2 \\ 6 \end{bmatrix} \right] \right] = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$